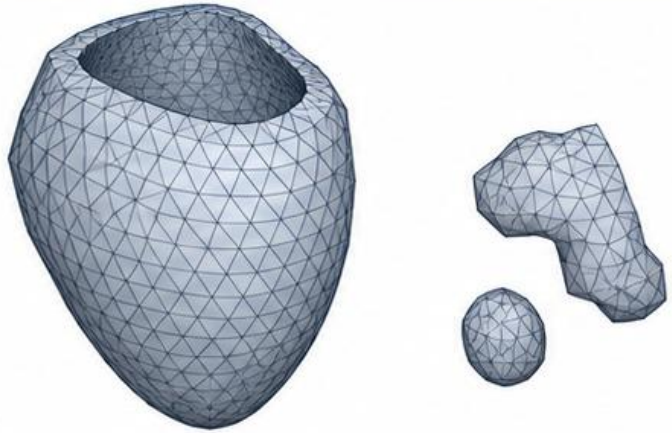
Cardiac Surface Split + CAMP

1. Find connected components and use the largest as the main shell; save detached components separately.
2. Generate candidate interior points around the mesh center of mass.
3. Score triangles by the orientation of their normals relative to each candidate point.
4. Choose coherent inner and outer seed regions, then assign remaining faces by shortest geodesic distance across face adjacency. This gives a curved cut that follows the mesh.
5. Remove the rim from the outer surface, where the wall folds back toward the cavity.
6. Check connectedness, boundary loops, matching cut edges, area balance, and seed coherence.

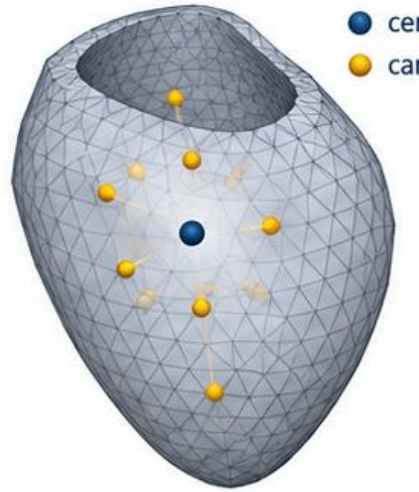
1 Largest component



Largest connected component
(closed triangulated shell)

Other mesh islands
(saved separately)

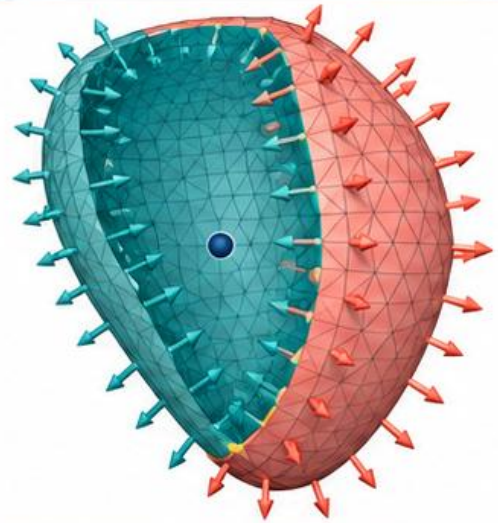
2 Candidate centers



center of mass

candidate points

3 Normal orientation

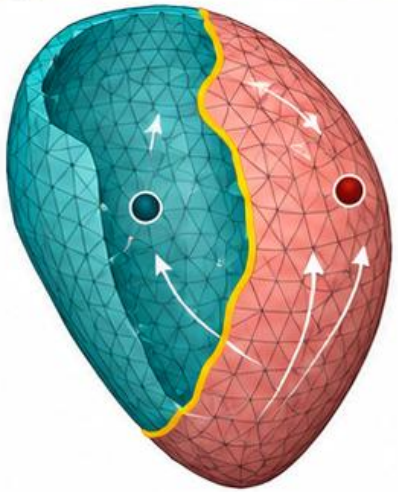


inward-facing triangles (score < 0)

outward-facing triangles (score > 0)

relative to candidate point

4 Geodesic assignment



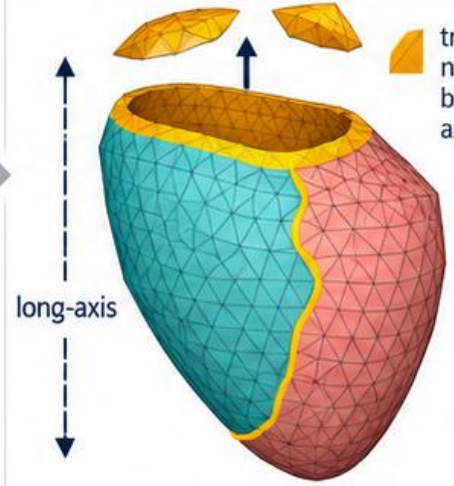
inner seed
(on inner surface)

outer seed
(on outer surface)

assign remaining faces by shortest paths on face-adjacency graph

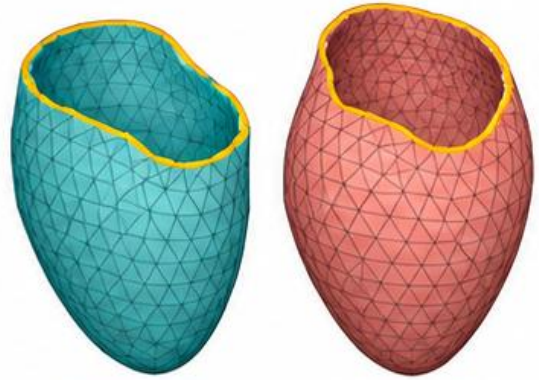
curved mesh seam, no manual plane

5 Basal rim trim



trim faces whose normals point toward base or back toward axis

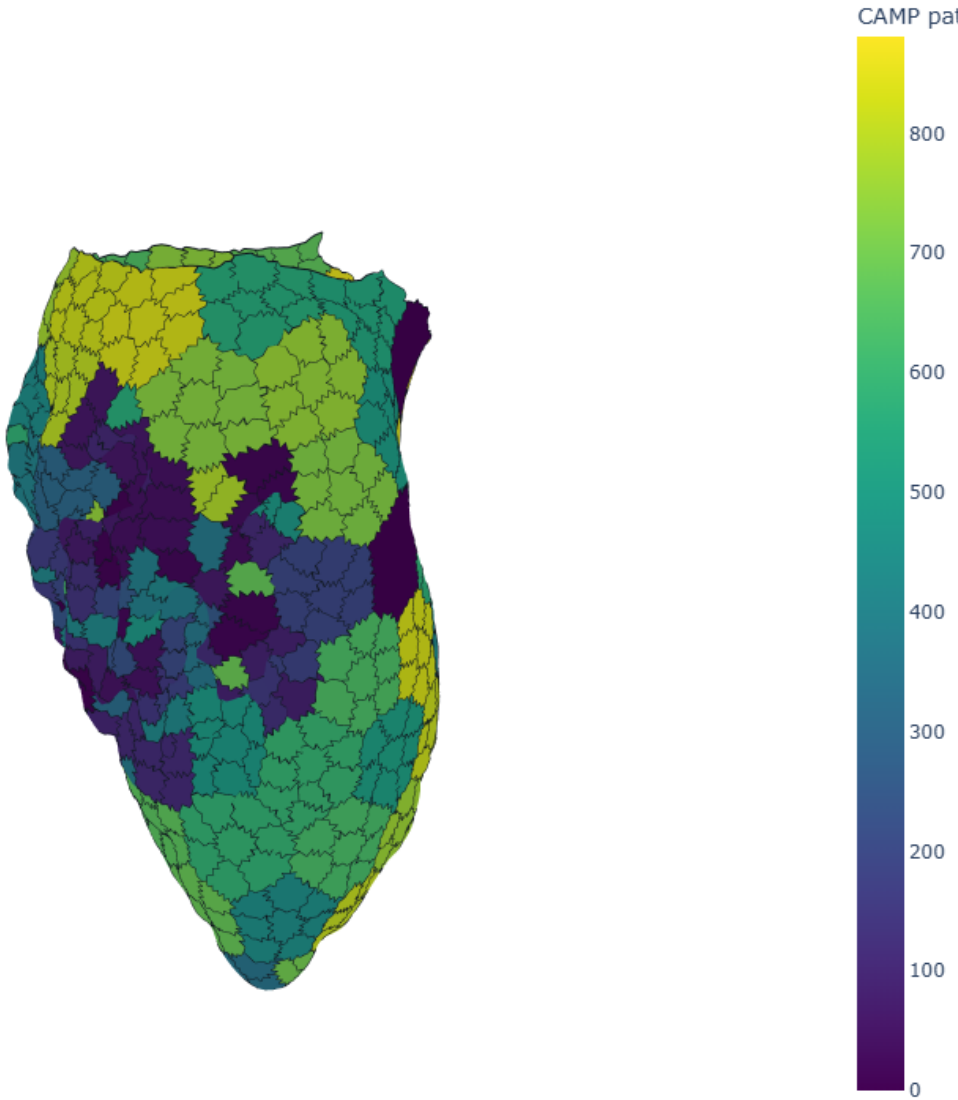
6 QC



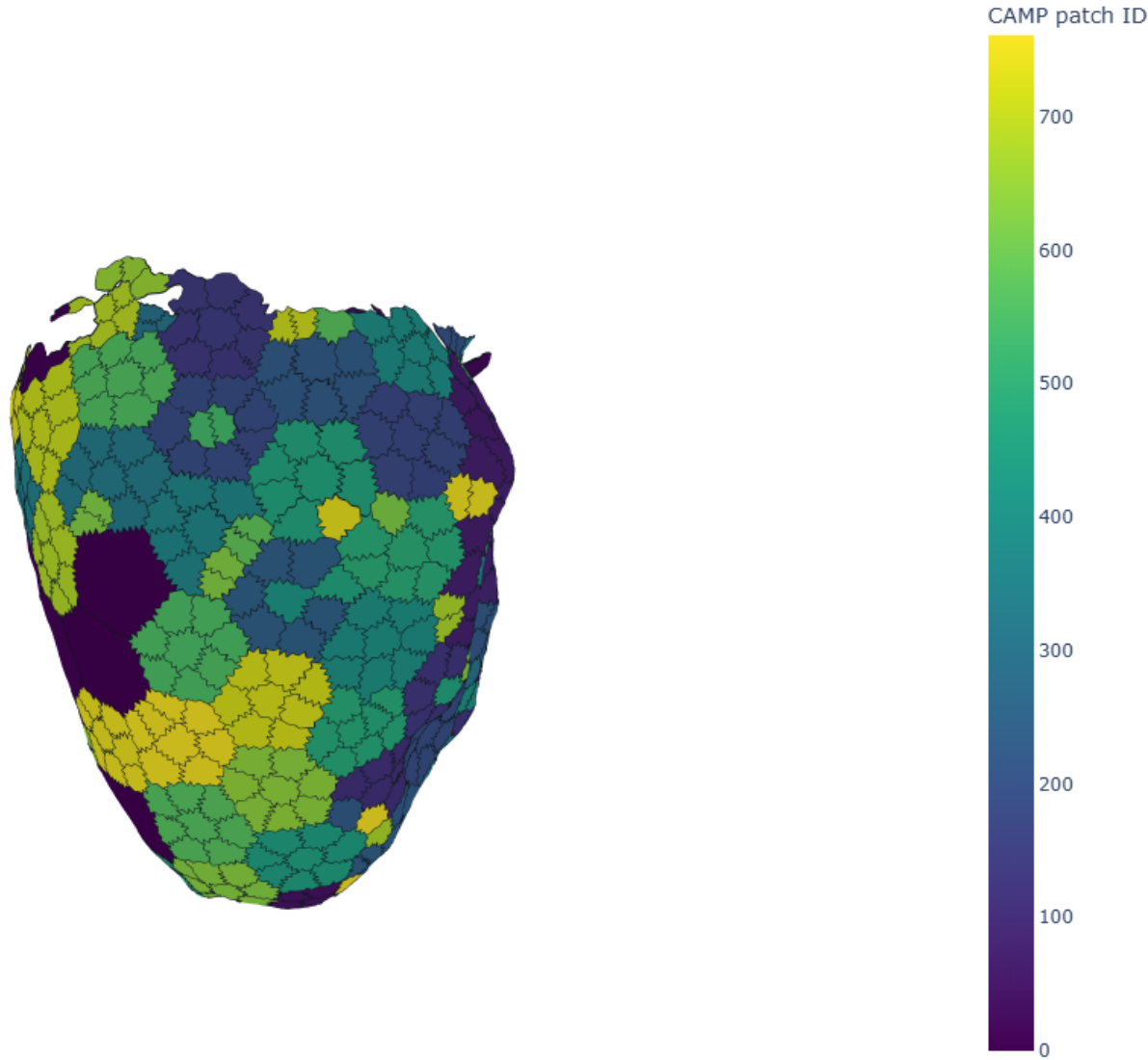
inner surface
(open, with boundary)

outer surface
(open, with boundary)

- ✓ connectedness
- ✓ simple boundary loops
- ✓ matching cut edges
- ✓ area balance
- ✓ seed coherence



Inner Surface



Outer Surface

Things I will be ironing out

1. Implement a method for handling holes in the segmented mesh.
2. Address the top-cut concern. The current splitter doesn’t use a manually placed cutting plane or the angle tolerance rule that Luka described.